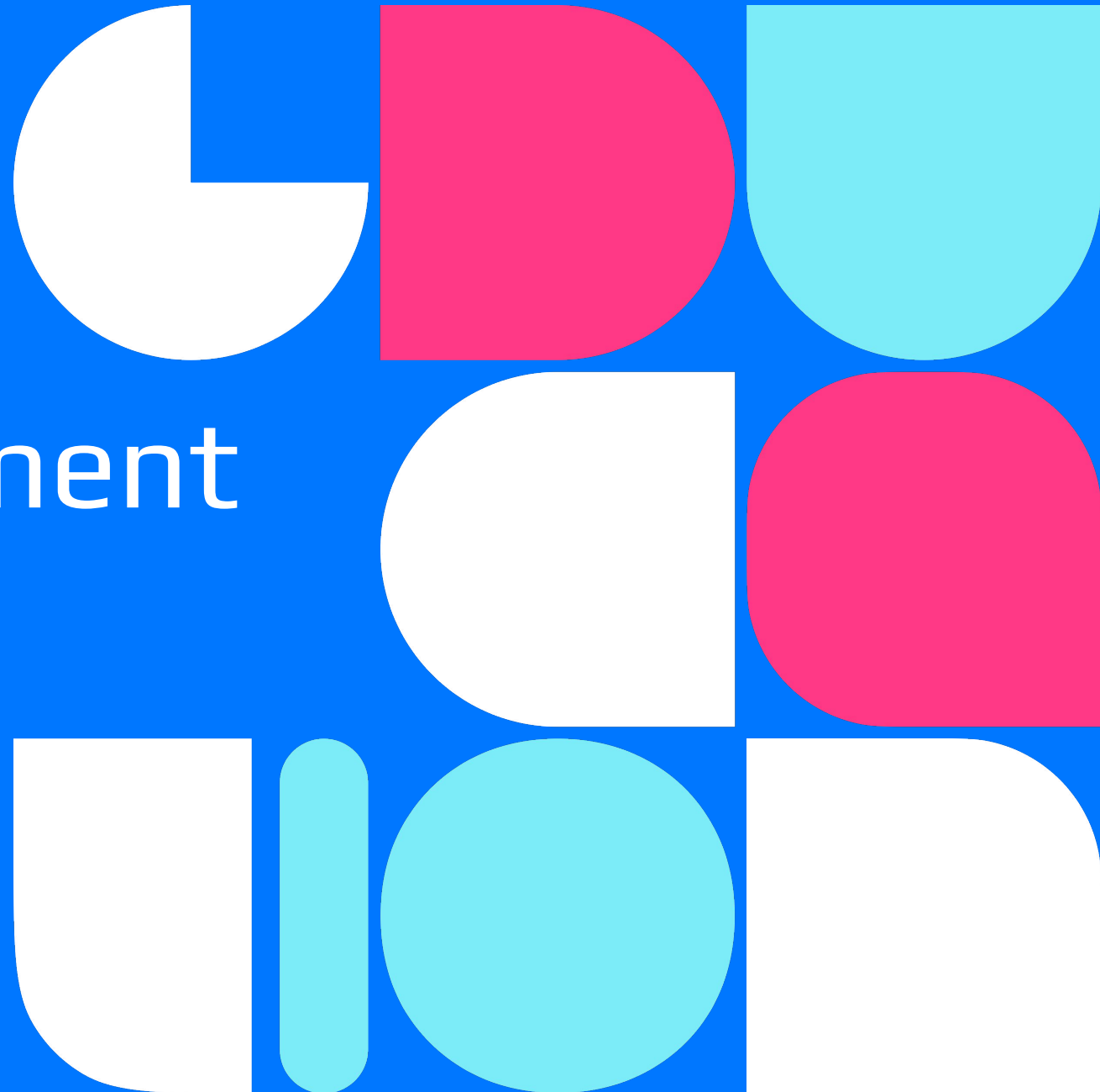


Лекция 4. Alignment

Елисеев Антон
05.03.2025



Instruction following



Instruction following

Paper: <https://arxiv.org/abs/2203.02155>

Prompt:

What is the purpose of the list C in the code below?

```
def binomial_coefficient(n, r):  
    C = [0 for i in range(r + 1)];  
    C[0] = 1;  
    for i in range(1, n + 1):  
        j = min(i, r);  
        while j > 0:  
            C[j] += C[j - 1];  
            j -= 1;  
    return C[r]
```

GPT-3 175B completion:

- A. to store the value of C[0]
- B. to store the value of C[1]
- C. to store the value of C[i]
- D. to store the value of C[i - 1]

InstructGPT 175B completion:

The list C in this code is used to store the values of the binomial coefficient as the function iterates through the values of n and r. It is used to calculate the value of the binomial coefficient for a given value of n and r, and stores the result in the final return value of the function.

Instruction following

Training objective:

what we want vs
what we told model to do

What we told model to do:

- predict the next token on a webpage from the internet

Alignment



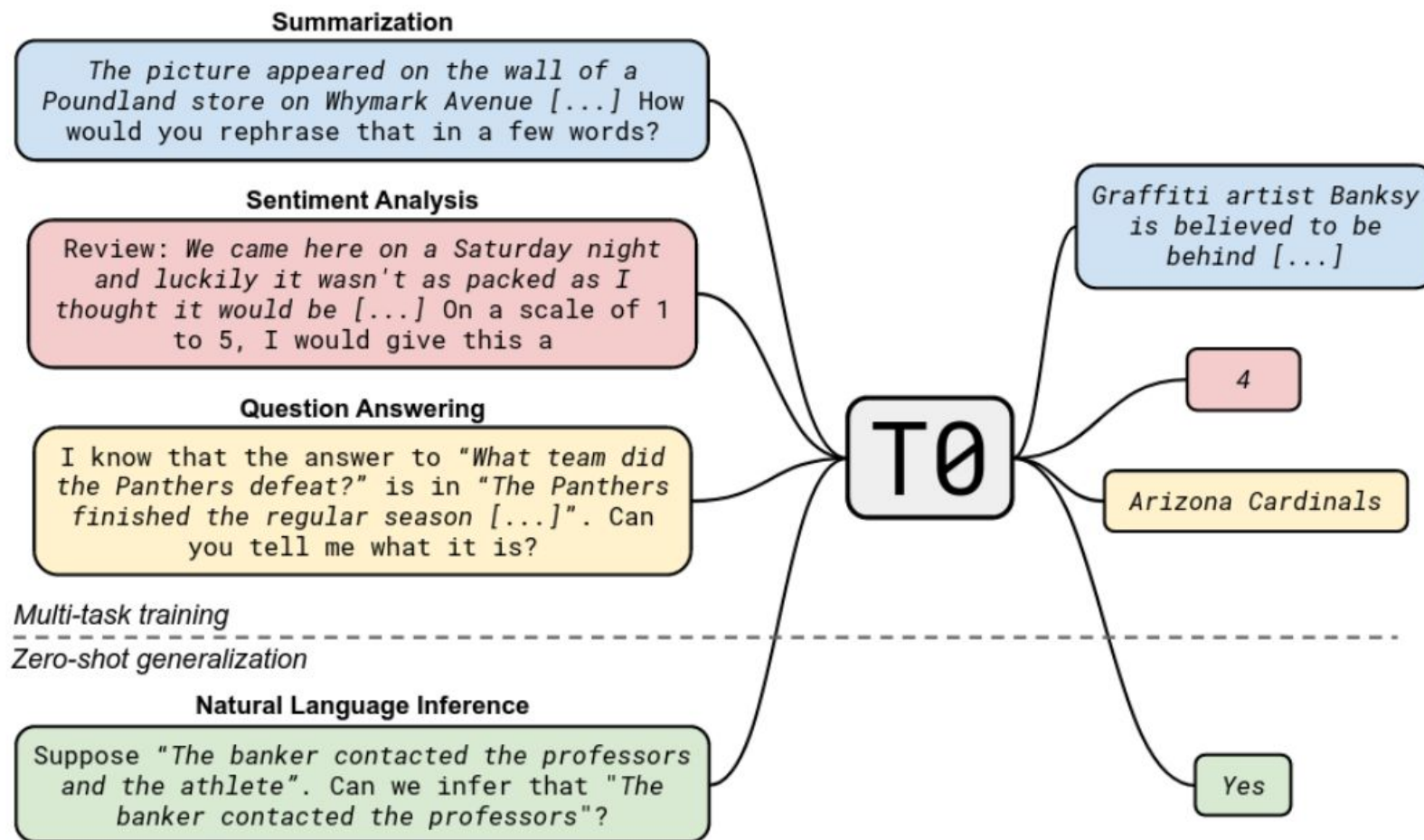
What we want model to do:

- follow the user's instructions helpfully and safely

The language modeling objective is misaligned

Instruction following

Paper: <https://arxiv.org/abs/2110.08207>



SFT issues



SFT issues

Вопрос: В чем проблемы SFT?

SFT issues

- Supervised seq2seq learning:

$$P(y_{t+1}|x, y_{0:t}), \quad y_{0:t} \sim \text{reference}$$

- Inference

$$P(y_{t+1}|x, \hat{y}_{0:t}), \quad \hat{y}_{0:t} \sim ???$$

SFT issues

- Supervised seq2seq learning:

$$P(y_{t+1}|x, y_{0:t}), \quad y_{0:t} \sim \text{reference}$$

- Inference

$$P(y_{t+1}|x, \hat{y}_{0:t}), \quad \hat{y}_{0:t} \sim \text{model}$$

**If model ever makes something that isn't in data,
It gets volatile from next time-step!**

SFT issues

There's more than one correct translation.
You don't need to learn all of them.

Source: 在 找 给 家 里 人 的 礼 物 .

Versions:

i 'm searching for some gifts for my family.
i want to find something for my family as presents.
i 'm about to buy some presents for my family.
i 'd like to buy my family something as a gift.
i 'm looking for a present for my family.
...

SFT issues

There's more than one correct translation.
You don't need to learn all of them.

Source: 在 找 给 家 里 人 的 礼 物 .

Versions:	Model 1 $p(y x)$	Model 2 $p(y x)$
(version 1)	1e-2	0.99
(version 2)	2e-2	1e-100
(version 3)	1e-2	1e-100
(all rubbish)	0.96	0.01

SFT issues

There's more than one correct translation.
You don't need to learn all of them.

Source: 在 找 给 家 里 人 的 礼 物 .

Versions:	Model 1 $p(y x)$	Model 2 $p(y x)$
(version 1)	1e-2	0.99
(version 2)	2e-2	1e-100
(version 3)	1e-2	1e-100
(all rubbish)	0.96	0.01
	better llh	worse llh
	96% rubbish	1% rubbish

SFT issues

Вопрос: Почему тогда не использовать только RL?

SFT issues

Вопрос: Почему тогда не использовать только RL?

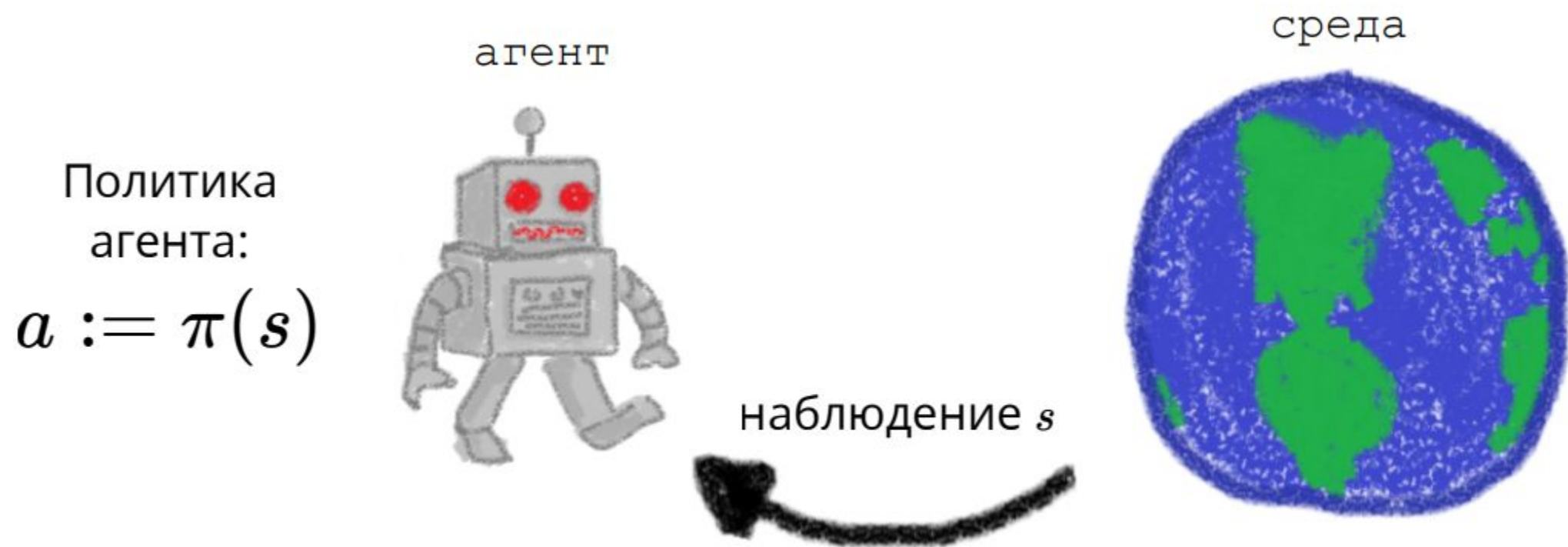
1. Это RL -> плохо сходится, нужно много данных
2. Говорит “плохо”, но не говорит, как правильно

Вывод: делаем SFT, поверх него RL

RL recap



RL recap



Decision Process - выбор действий по наблюдениям

RL recap

Supervised learning:

$$\nabla llh = E_{s, a_{opt} \sim D} \nabla \log \pi_{\theta}(a_{opt}|s)$$

Policy gradient:

$$\nabla J = E_{\substack{s \sim d(s) \\ a \sim \pi(a|obs(s))}} \nabla \log \pi_{\theta}(a|s) Q(s, a)$$

RL recap

Supervised learning:

$$\nabla llh = E_{s, a_{opt} \sim D} \nabla \log \pi_{\theta}(a_{opt}|s)$$

Policy gradient:

reference

$$\nabla J = E_{\substack{s \sim d(s) \\ a \sim \pi(a|obs(s))}} \nabla \log \pi_{\theta}(a|s) Q(s, a)$$

generated

RLHF



RLHF

Paper: <https://arxiv.org/abs/2203.02155>

Step 1

**Collect demonstration data,
and train a supervised policy.**

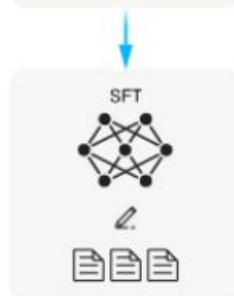
A prompt is
sampled from our
prompt dataset.



A labeler
demonstrates the
desired output
behavior.



This data is used
to fine-tune GPT-3
with supervised
learning.



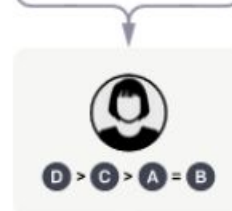
Step 2

**Collect comparison data,
and train a reward model.**

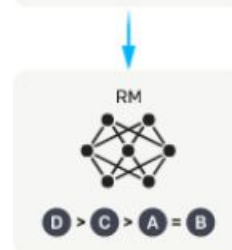
A prompt and
several model
outputs are
sampled.



A labeler ranks
the outputs from
best to worst.



This data is used
to train our
reward model.



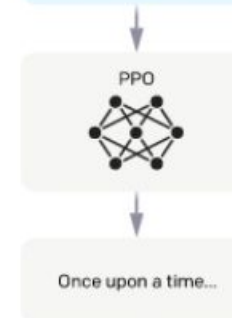
Step 3

**Optimize a policy against
the reward model using
reinforcement learning.**

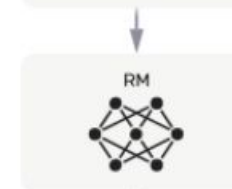
A new prompt
is sampled from
the dataset.



The policy
generates
an output.



The reward model
calculates a
reward for
the output.



The reward is
used to update
the policy
using PPO.



RLHF

$$\mathbf{E}_{a \sim \pi_{\theta}(a|s)} \left[r_{\psi}(s, a) - \beta \text{KL}(\pi_{\theta}(a|s) || \pi_{\text{SFT}}(a|s)) \right] \rightarrow \max_{\theta}$$

RLHF

Paper: <https://arxiv.org/abs/2203.02155>

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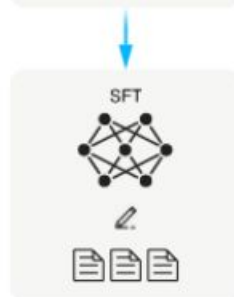
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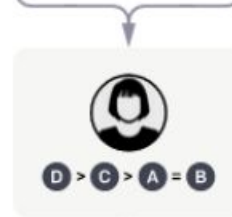
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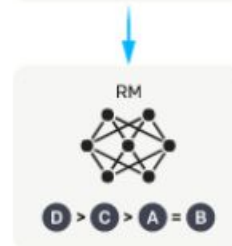
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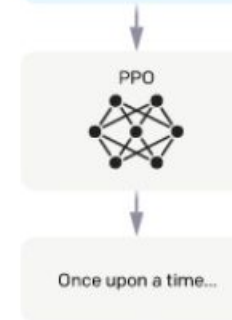
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The reward is
used to update
the policy
using PPO.



Reward modeling



Reward modeling. Модель Брэдли-Терри

Вероятностная модель: $P(a > b|s) = \sigma(r_\psi(s, a) - r_\psi(s, b))$

Максимизируем правдоподобие: $\sum_{(s, winner, loser) \in \mathbf{D}} \log \sigma(r_\psi(s, winner) - r_\psi(s, loser)) \rightarrow \max_{\psi}$

RLHF

Paper: <https://arxiv.org/abs/2203.02155>

Step 1

**Collect demonstration data,
and train a supervised policy.**

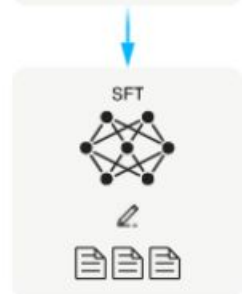
A prompt is
sampled from our
prompt dataset.



A labeler
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desired output
behavior.



This data is used
to fine-tune GPT-3
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Step 2

**Collect comparison data,
and train a reward model.**

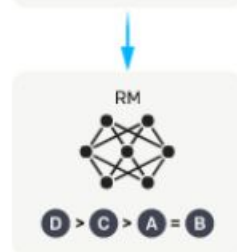
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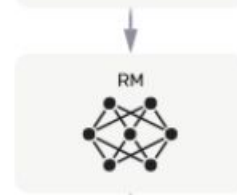
A new prompt
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The reward is
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using PPO.



PPO issues



PPO issues

Вопрос: Какие проблемы у такого подхода?

PPO issues

Вопрос: Какие проблемы у такого подхода?

1. Это RL -> плохо сходится, у PPO много гиперпараметров
2. Много GPU памяти
Пример: gpt3 175B требует 7TB gpu-памяти в BF16 ~ 88 A100 80G ~ \$2M
3. Надо инферить модель на каждом шагу

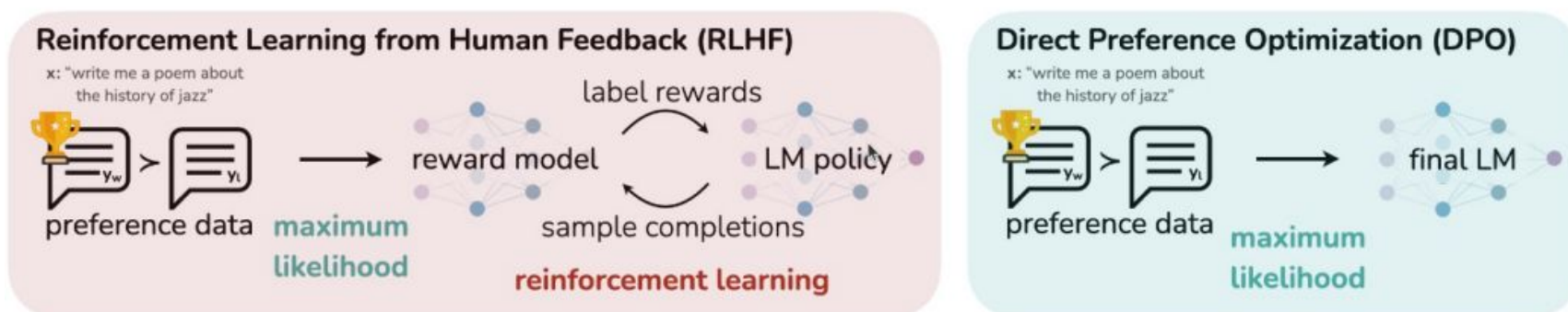
DPO



DPO

Paper: <https://arxiv.org/abs/2305.18290>

RLHF is complicated, let's make it simpler!



Direct Preference Optimization: Your Language Model is Secretly a Reward Model

Rafael Rafailov, Archit Sharma, Eric Mitchell, Stefano Ermon, Christopher D. Manning, Chelsea Finn

While large-scale unsupervised language models (LMs) learn broad world knowledge and some reasoning skills, achieving precise control of their behavior methods for gaining such steerability collect human labels of the relative quality of model generations and fine-tune the unsupervised LM to align with human preferences. However, RLHF is a complex and often unstable procedure, first fitting a reward model that reflects the human preferences, and then fine-tuning the LM to maximize the reward without drifting too far from the original model. In this paper we introduce a new parameterization of the reward model in RLHF that enables us to solve the standard RLHF problem with only a simple classification loss. The resulting algorithm, which we call Direct Preference Optimization (DPO), is stable, precise, and does not require the LM during fine-tuning or performing significant hyperparameter tuning. Our experiments show that DPO can fine-tune LMs to align with human preferences and exceeds PPO-based RLHF in ability to control sentiment of generations, and matches or improves response quality in summarization and single-turn dialog.

DPO

$$\mathbf{E}_{a \sim \pi_{\theta}(a|s)} \left[r_{\psi}(s, a) - \beta \text{KL}(\pi_{\theta}(a|s) || \pi_{\text{SFT}}(a|s)) \right] \rightarrow \max_{\theta}$$

DPO

$$\mathbf{E}_{a \sim \pi_{\theta}(a|s)} \left[r_{\psi}(s, a) - \beta \text{KL}(\pi_{\theta}(a|s) || \pi_{\text{SFT}}(a|s)) \right] \rightarrow \max_{\theta}$$

$$\pi^*(a|s) = \frac{1}{Z(s)} \pi_{\text{SFT}}(a|s) e^{\frac{1}{\beta} r(s,a)}$$

$$\text{where } Z(s) = \sum_a e^{\frac{1}{\beta} r(s,a)}$$

DPO

$$\pi^*(a|s) = \frac{1}{Z(s)} \pi_{\text{SFT}}(a|s) e^{\frac{1}{\beta} r(s,a)} \quad \longrightarrow \quad r(s, a) = \beta \log \frac{\pi^*(a|s)}{\pi_{\text{SFT}}(a|s)} + \beta \log Z(s)$$

$$Z(s) = \sum_a e^{\frac{1}{\beta} r(s,a)}$$

$$r_{\theta}(s, a) = \beta \log \frac{\pi_{\theta}(a|s)}{\pi_{\text{SFT}}(a|s)} + \beta \log Z(s)$$

DPO

$$\sum_{(s, \text{winner}, \text{loser}) \in \mathbf{D}} \log \sigma(r_{\theta}(s, \text{winner}) - r_{\theta}(s, \text{loser})) \rightarrow \max_{\theta}$$

$$r_{\theta}(s, a) = \beta \log \frac{\pi_{\theta}(a|s)}{\pi_{\text{SFT}}(a|s)} + \beta \log Z(s)$$

$$\sum_{(s, \text{winner}, \text{loser}) \in \mathbf{D}} \log \sigma\left(\beta \left[\log \frac{\pi_{\theta}(\text{winner}|s)}{\pi_{\text{SFT}}(\text{winner}|s)} - \log \frac{\pi_{\theta}(\text{loser}|s)}{\pi_{\text{SFT}}(\text{loser}|s)} \right]\right) \rightarrow \max_{\theta}$$

DPO

Плюсы:

1. не RL -> лучше и быстрее сходится
2. Только один гиперпараметр - β
3. Надо хранить только обучаемую модель
4. Не инферим, только прогоняем существующий ответ

DPO variations

Method	Objective
RRHF [91]	$\max \left(0, -\frac{1}{ y_w } \log \pi_\theta(y_w x) + \frac{1}{ y_l } \log \pi_\theta(y_l x) \right) - \lambda \log \pi_\theta(y_w x)$
SLiC-HF [96]	$\max (0, \delta - \log \pi_\theta(y_w x) + \log \pi_\theta(y_l x)) - \lambda \log \pi_\theta(y_w x)$
DPO [66]	$-\log \sigma \left(\beta \log \frac{\pi_\theta(y_w x)}{\pi_{\text{ref}}(y_w x)} - \beta \log \frac{\pi_\theta(y_l x)}{\pi_{\text{ref}}(y_l x)} \right)$
IPO [6]	$\left(\log \frac{\pi_\theta(y_w x)}{\pi_{\text{ref}}(y_w x)} - \log \frac{\pi_\theta(y_l x)}{\pi_{\text{ref}}(y_l x)} - \frac{1}{2\tau} \right)^2$
CPO [88]	$-\log \sigma (\beta \log \pi_\theta(y_w x) - \beta \log \pi_\theta(y_l x)) - \lambda \log \pi_\theta(y_w x)$
KTO [29]	$-\lambda_w \sigma \left(\beta \log \frac{\pi_\theta(y_w x)}{\pi_{\text{ref}}(y_w x)} - z_{\text{ref}} \right) + \lambda_l \sigma \left(z_{\text{ref}} - \beta \log \frac{\pi_\theta(y_l x)}{\pi_{\text{ref}}(y_l x)} \right),$ where $z_{\text{ref}} = \mathbb{E}_{(x,y) \sim \mathcal{D}} [\beta \text{KL}(\pi_\theta(y x) \pi_{\text{ref}}(y x))]$
ORPO [42]	$-\log p_\theta(y_w x) - \lambda \log \sigma \left(\log \frac{p_\theta(y_w x)}{1-p_\theta(y_w x)} - \log \frac{p_\theta(y_l x)}{1-p_\theta(y_l x)} \right),$ where $p_\theta(y x) = \exp \left(\frac{1}{ y } \log \pi_\theta(y x) \right)$
R-DPO [64]	$-\log \sigma \left(\beta \log \frac{\pi_\theta(y_w x)}{\pi_{\text{ref}}(y_w x)} - \beta \log \frac{\pi_\theta(y_l x)}{\pi_{\text{ref}}(y_l x)} + (\alpha y_w - \alpha y_l) \right)$
SimPO	$-\log \sigma \left(\frac{\beta}{ y_w } \log \pi_\theta(y_w x) - \frac{\beta}{ y_l } \log \pi_\theta(y_l x) - \gamma \right)$

SimPO



SimPO (NeurIPS 2024)

$$\mathcal{L}_{\text{DPO}}(\pi_{\theta}; \pi_{\text{ref}}) = -\mathbb{E}_{(x, y_w, y_l) \sim \mathcal{D}} \left[\log \sigma \left(\beta \log \frac{\pi_{\theta}(y_w | x)}{\pi_{\text{ref}}(y_w | x)} - \beta \log \frac{\pi_{\theta}(y_l | x)}{\pi_{\text{ref}}(y_l | x)} \right) \right]$$

Проблемы в DPO

- нужна SFT модель в памяти
- оптимизируем не то, что хотим на инференсе

SimPO (NeurIPS 2024)

$$-\mathbb{E}_{(x, y_w, y_l) \sim \mathcal{D}} \left[\log \sigma \left(\beta \log \frac{\pi_{\theta}(y_w | x)}{\pi_{\theta}(y_l | x)} \right) \right]$$

SimPO (NeurIPS 2024)

$$r_{\text{SimPO}}(x, y) = \frac{\beta}{|y|} \log \pi_{\theta}(y \mid x) = \frac{\beta}{|y|} \sum_{i=1}^{|y|} \log \pi_{\theta}(y_i \mid x, y_{<i})$$

SimPO (NeurIPS 2024)

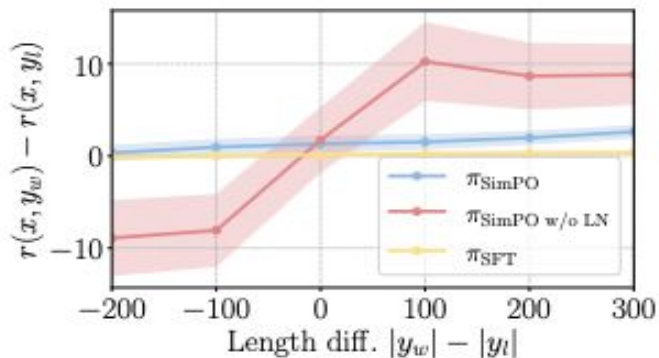
$$p(y_w \succ y_l \mid x) = \sigma(r(x, y_w) - r(x, y_l) - \gamma)$$

$$\mathcal{L}_{\text{SimPO}}(\pi_\theta) = -\mathbb{E}_{(x, y_w, y_l) \sim \mathcal{D}} \left[\log \sigma \left(\frac{\beta}{|y_w|} \log \pi_\theta(y_w | x) - \frac{\beta}{|y_l|} \log \pi_\theta(y_l | x) - \gamma \right) \right]$$

SimPO (NeurIPS 2024)

Method	Mistral-Base (7B) Setting					Mistral-Instruct (7B) Setting				
	AlpacaEval 2		Arena-Hard	MT-Bench		AlpacaEval 2		Arena-Hard	MT-Bench	
	LC (%)	WR (%)	WR (%)	GPT-4 Turbo	GPT-4	LC (%)	WR (%)	WR (%)	GPT-4 Turbo	GPT-4
DPO	15.1	12.5	10.4	5.9	7.3	26.8	24.9	16.3	6.3	7.6
SimPO	21.5	20.8	16.6	6.0	7.3	32.1	34.8	21.0	6.6	7.6
w/o LN	11.9	13.2	9.4	5.5	7.3	19.1	19.7	16.3	6.4	7.6
$\gamma = 0$	16.8	14.3	11.7	5.6	6.9	30.9	34.2	20.5	6.6	7.7

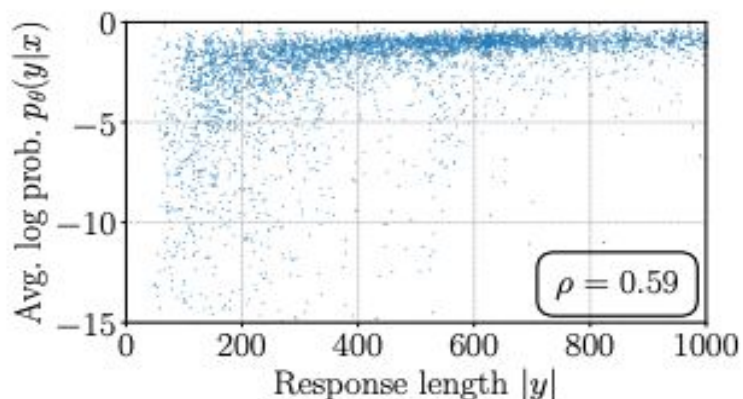
SimPO (NeurIPS 2024)



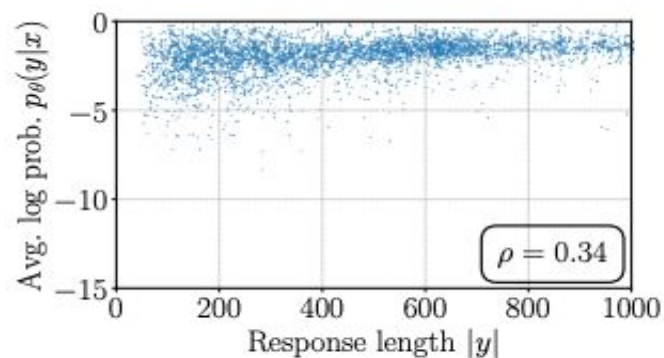
(a) Reward optimization.

Table 6: Spearman correlation ρ between average log likelihood of different models and response length on a held-out set.

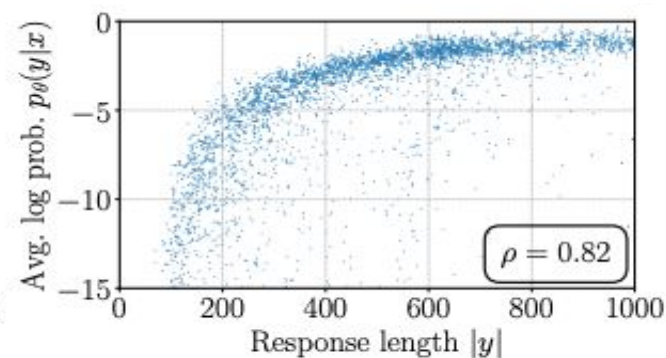
	SimPO w/o LN	DPO	SimPO
ρ	0.82	0.59	0.34



(a) Length correlation (DPO).



(b) SimPO.

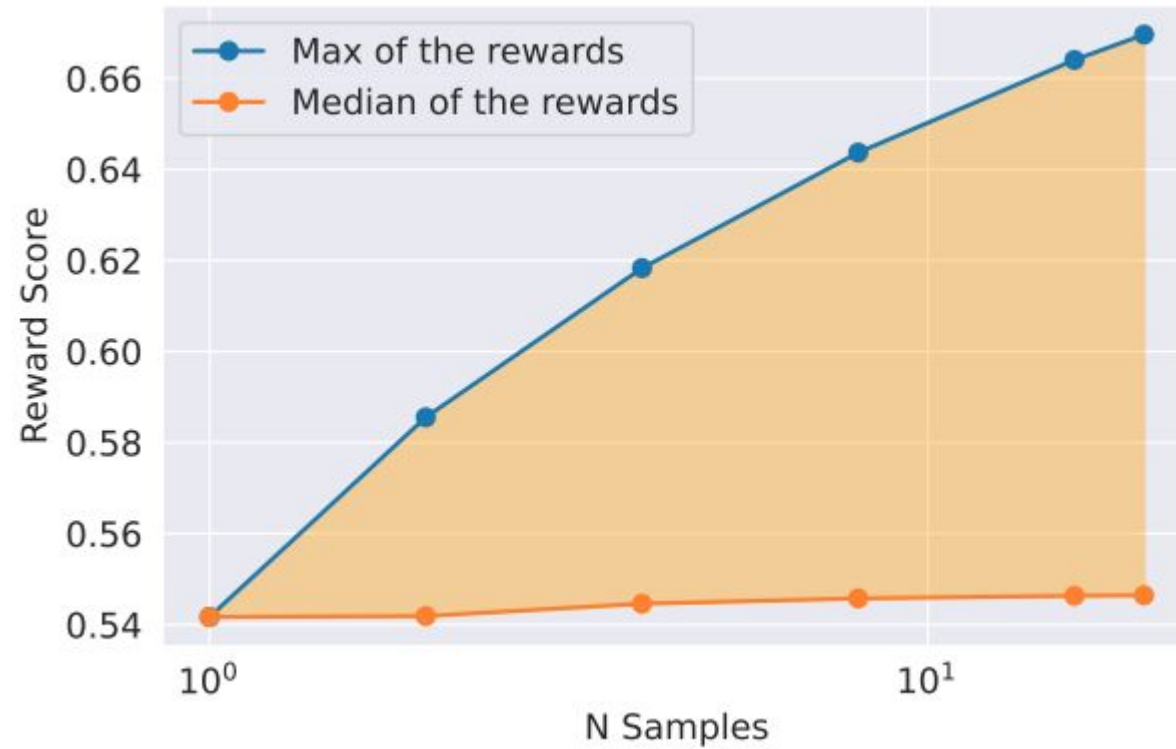


(c) SimPO without LN.

Modern approaches

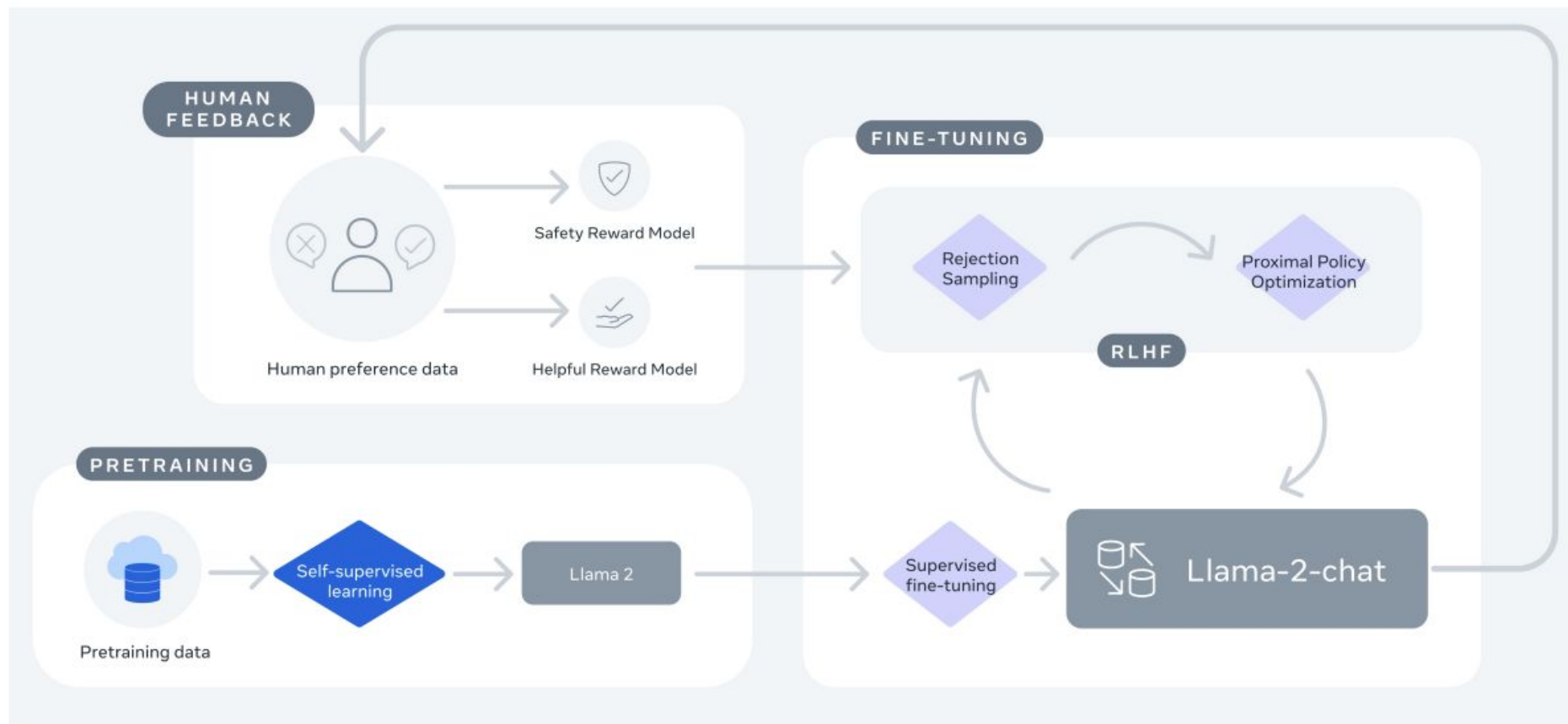


Rejection sampling



Llama 2

arXiv:2307.09288



Self-Rewarding Language Models

arXiv:2401.10020

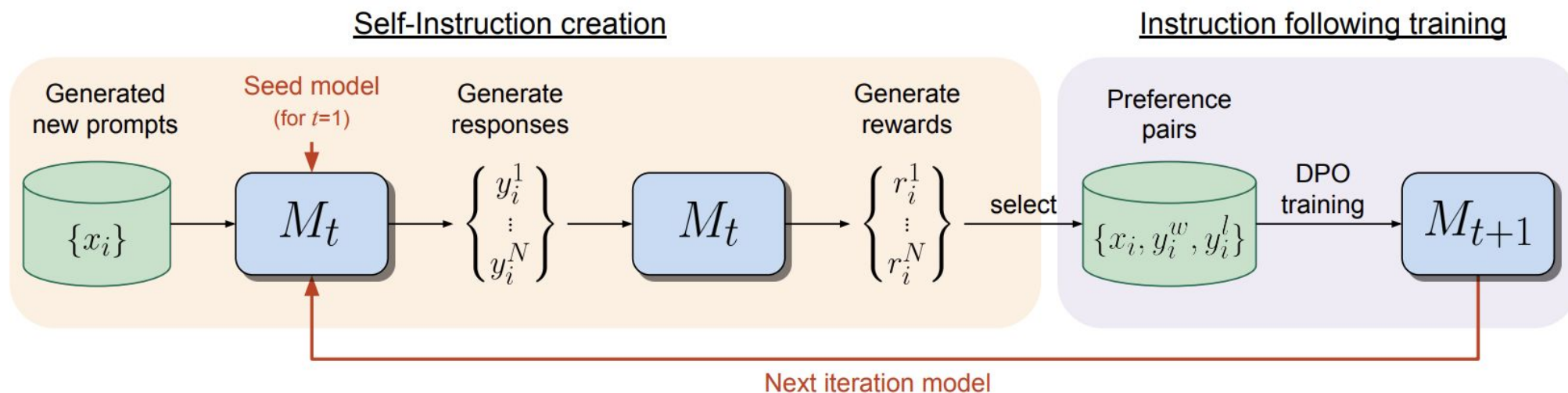


Figure 1: **Self-Rewarding Language Models.** Our self-alignment method consists of two steps: (i) *Self-Instruction creation*: newly created prompts are used to generate candidate responses from model M_t , which also predicts its own rewards via LLM-as-a-Judge prompting. (ii) *Instruction following training*: preference pairs are selected from the generated data, which are used for training via DPO, resulting in model M_{t+1} . This whole procedure can then be iterated resulting in both improved instruction following and reward modeling ability.

The Llama 3 Herd of Models

arXiv:2407.21783

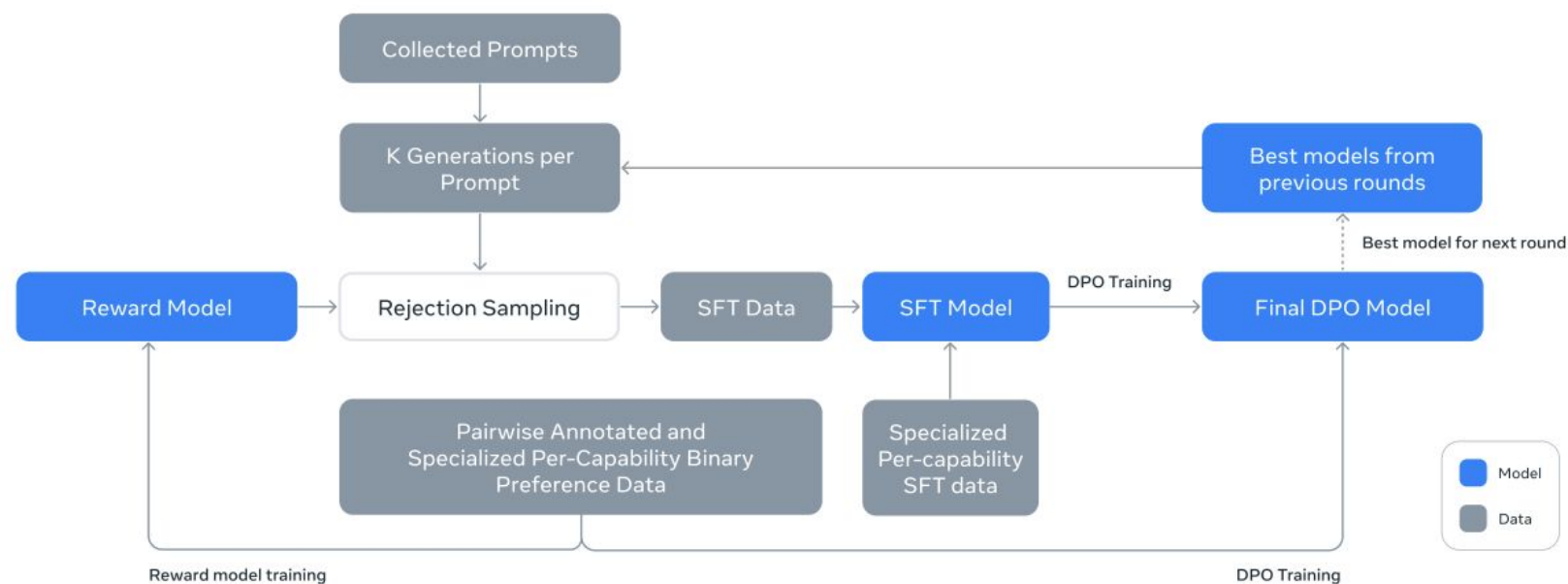


Figure 7 Illustration of the overall post-training approach for Llama 3. Our post-training strategy involves rejection sampling, supervised finetuning, and direct preference optimization. See text for details.

Data collection



Ultrachat (Enhancing Chat Language Models by Scaling High-quality Instructional Conversations)

arXiv:2305.14233

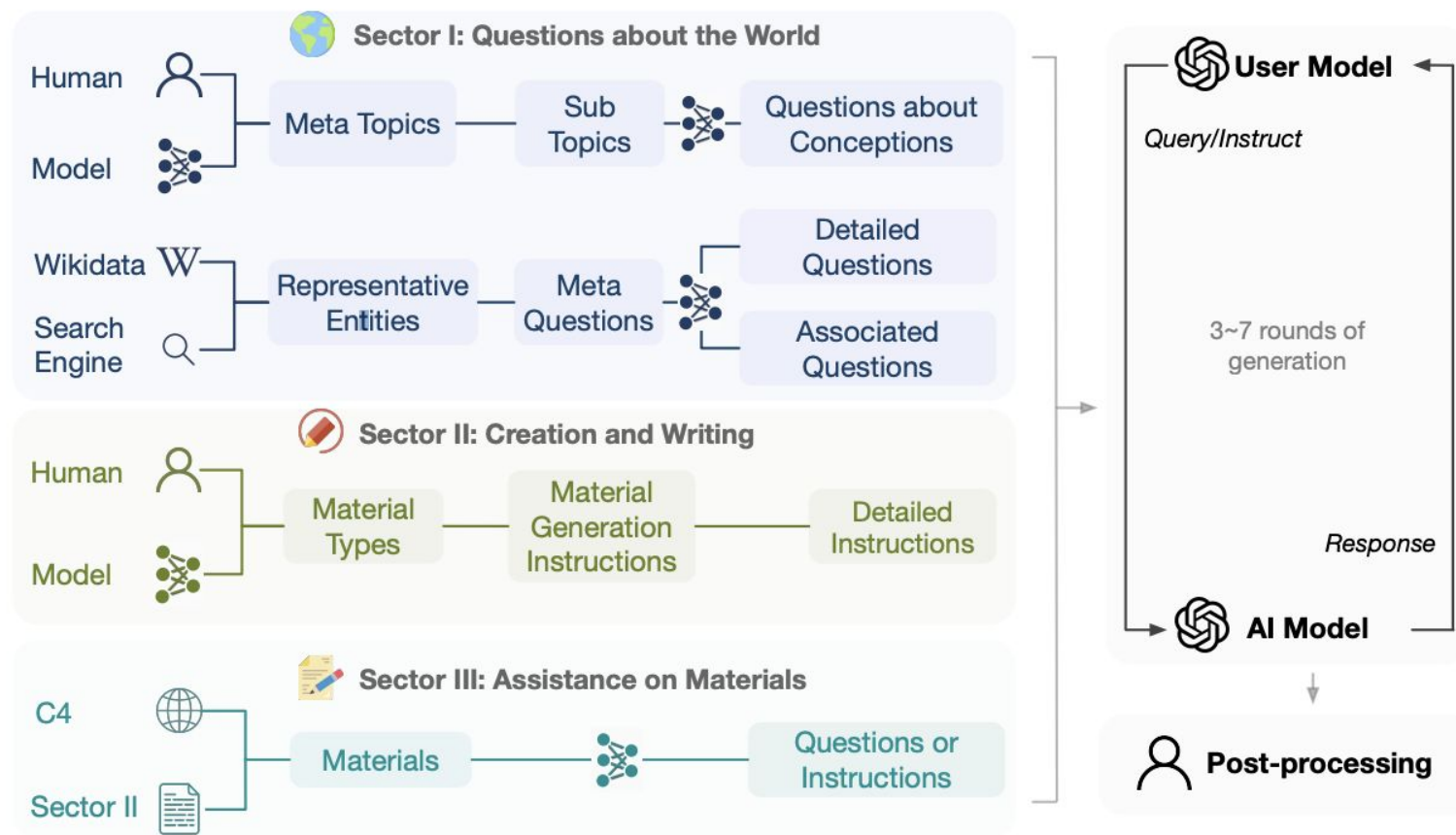
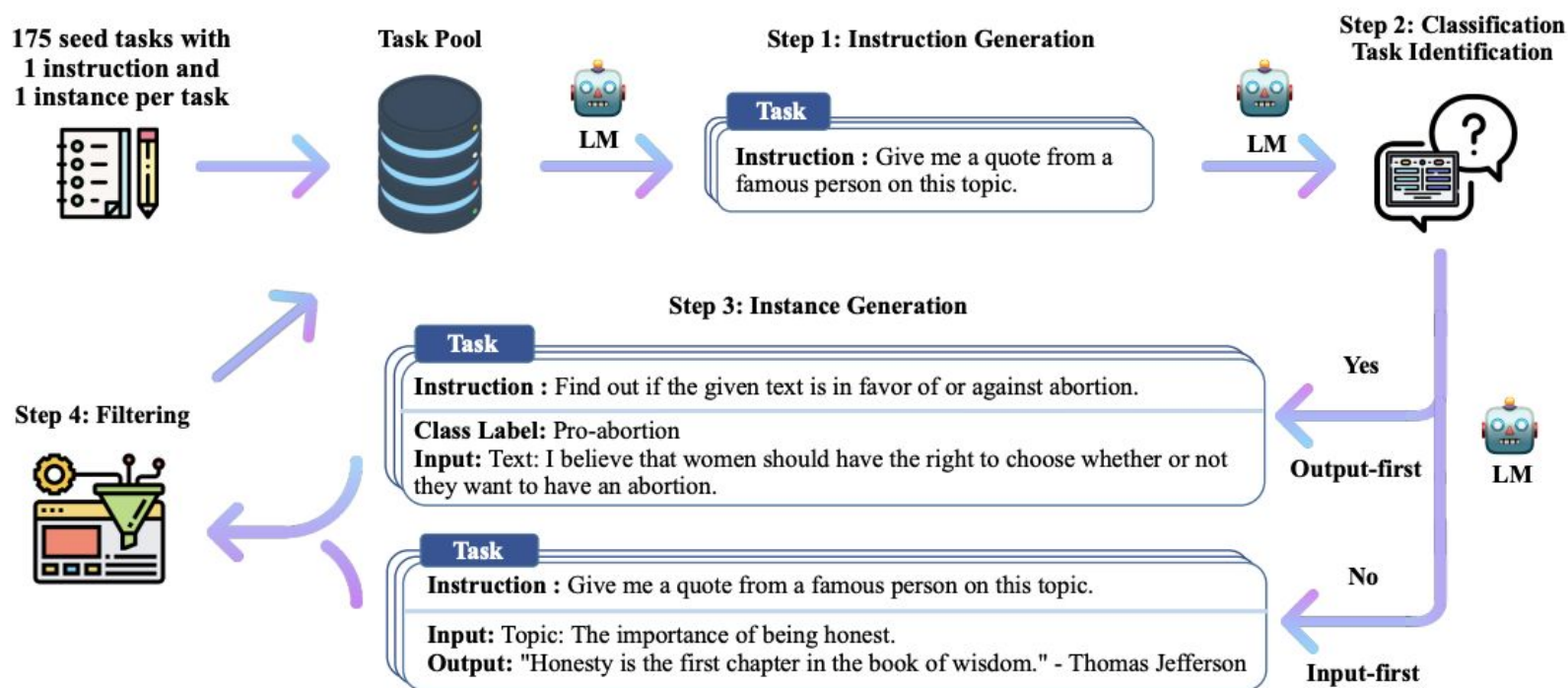


Figure 1: Construction process of UltraChat. The three sectors of data are derived from different meta information.

Self-Instruct: Aligning Language Models with Self-Generated Instructions

arXiv:2212.10560



Self-Instruct: Aligning Language Models with Self-Generated Instructions

arXiv:2212.10560

Come up with a series of tasks:

Task 1: {instruction for existing task 1}
Task 2: {instruction for existing task 2}
Task 3: {instruction for existing task 3}
Task 4: {instruction for existing task 4}
Task 5: {instruction for existing task 5}
Task 6: {instruction for existing task 6}
Task 7: {instruction for existing task 7}
Task 8: {instruction for existing task 8}
Task 9:

Evol-instruct

arXiv:2304.12244

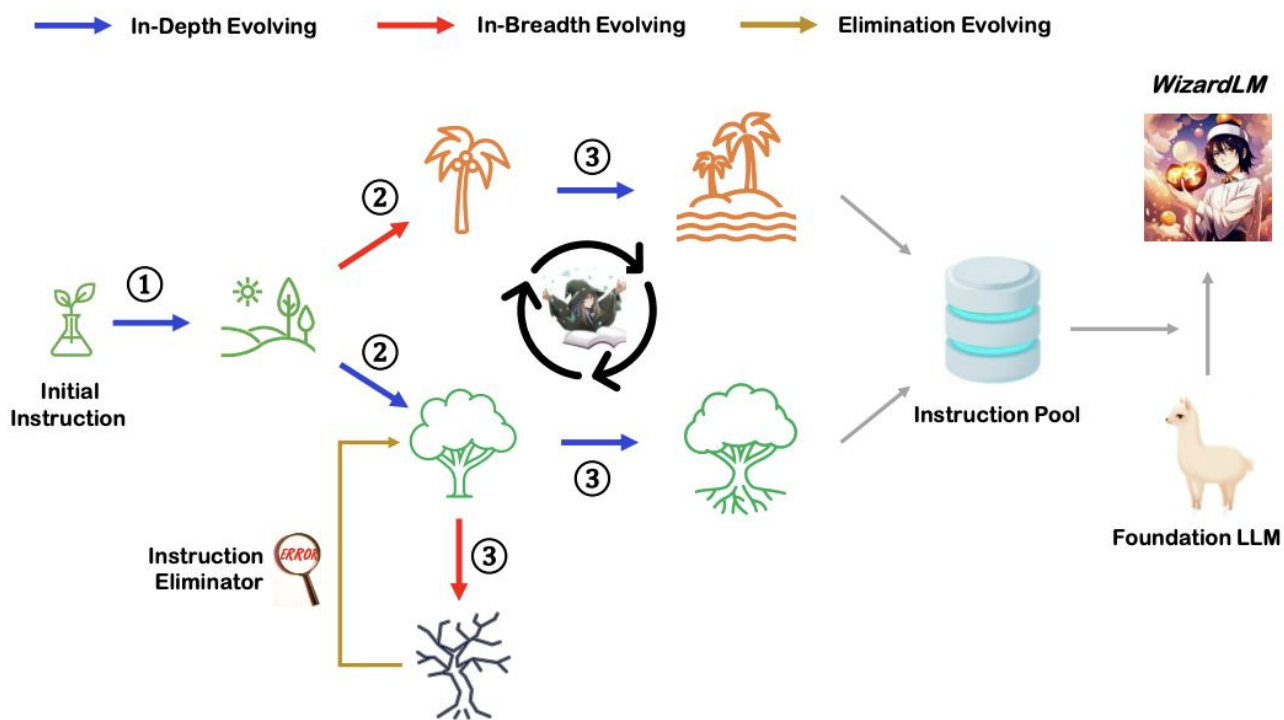
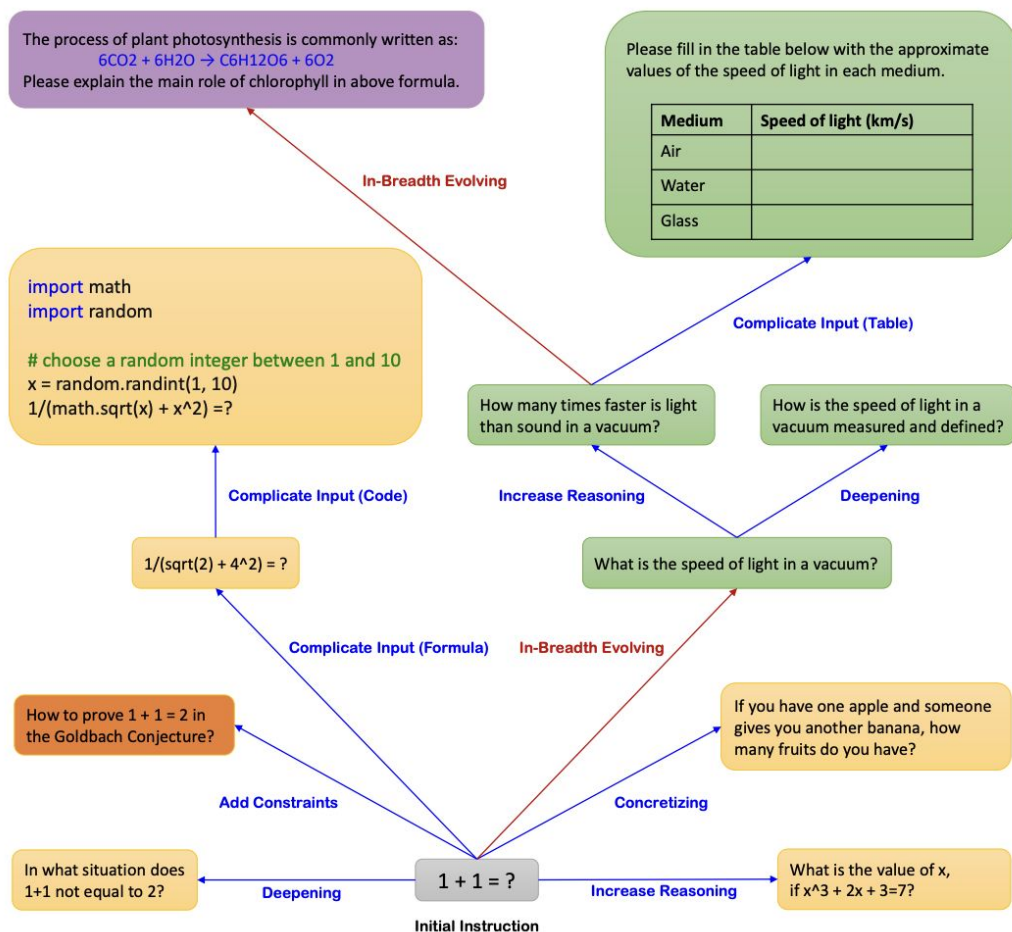


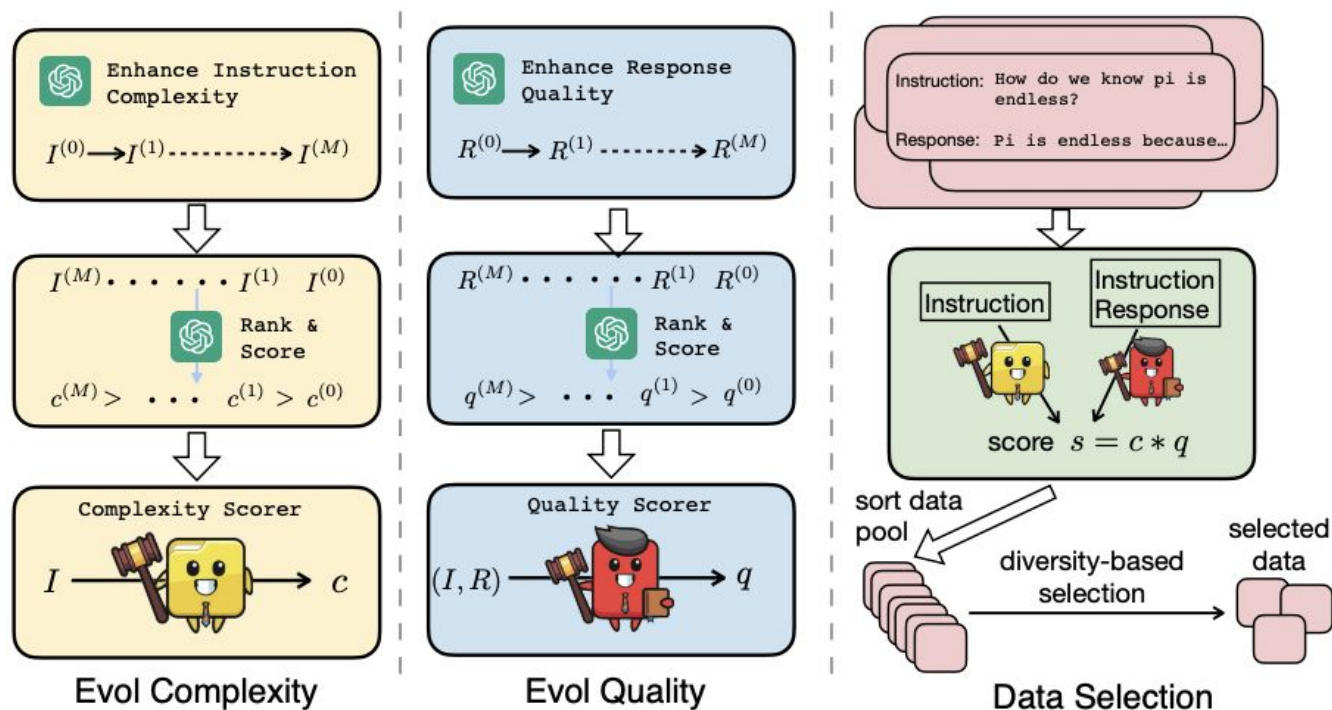
Figure 2: Overview of *Evol-Instruct*

Data filtering



What Makes Good Data for Alignment? A Comprehensive Study of Automatic Data Selection in Instruction Tuning

arXiv:2312.15685



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Algorithm 1 Score-First, Diversity-Aware Data Selection

```
1: Input: The data pool  $X$ , data budget  $m$ 
2: Output: The selected subset  $S_{\pi_{\text{DEITA}}}^{(m)}$ 
3: Initialize Empty Dataset  $S_{\pi_{\text{DEITA}}}^{(m)}$ 
4: Sorting  $X$  with the combined complexity score and quality score  $s = q * c$ ;
5: Getting the sorted Pool  $X^*$ ;
6: for Each Sample  $x \in X^*$  do
7:   //  $d(x, S)$  denotes the distance between  $x$  and its nearest neighbor in  $S$ 
8:   if  $d(x, S_{\pi_{\text{DEITA}}}^{(m)}) < \tau$  then
9:      $S_{\pi_{\text{DEITA}}}^{(m)} \leftarrow S_{\pi_{\text{DEITA}}}^{(m)} \cup \{x\}$ 
10:  else
11:    Continue
12:  end if
13:   $X \leftarrow X \setminus \{x\}$ 
14:  if  $|S_{\pi_{\text{DEITA}}}^{(m)}|$  equals to  $m$  then
15:    Break
16:  end if
17: end for
```

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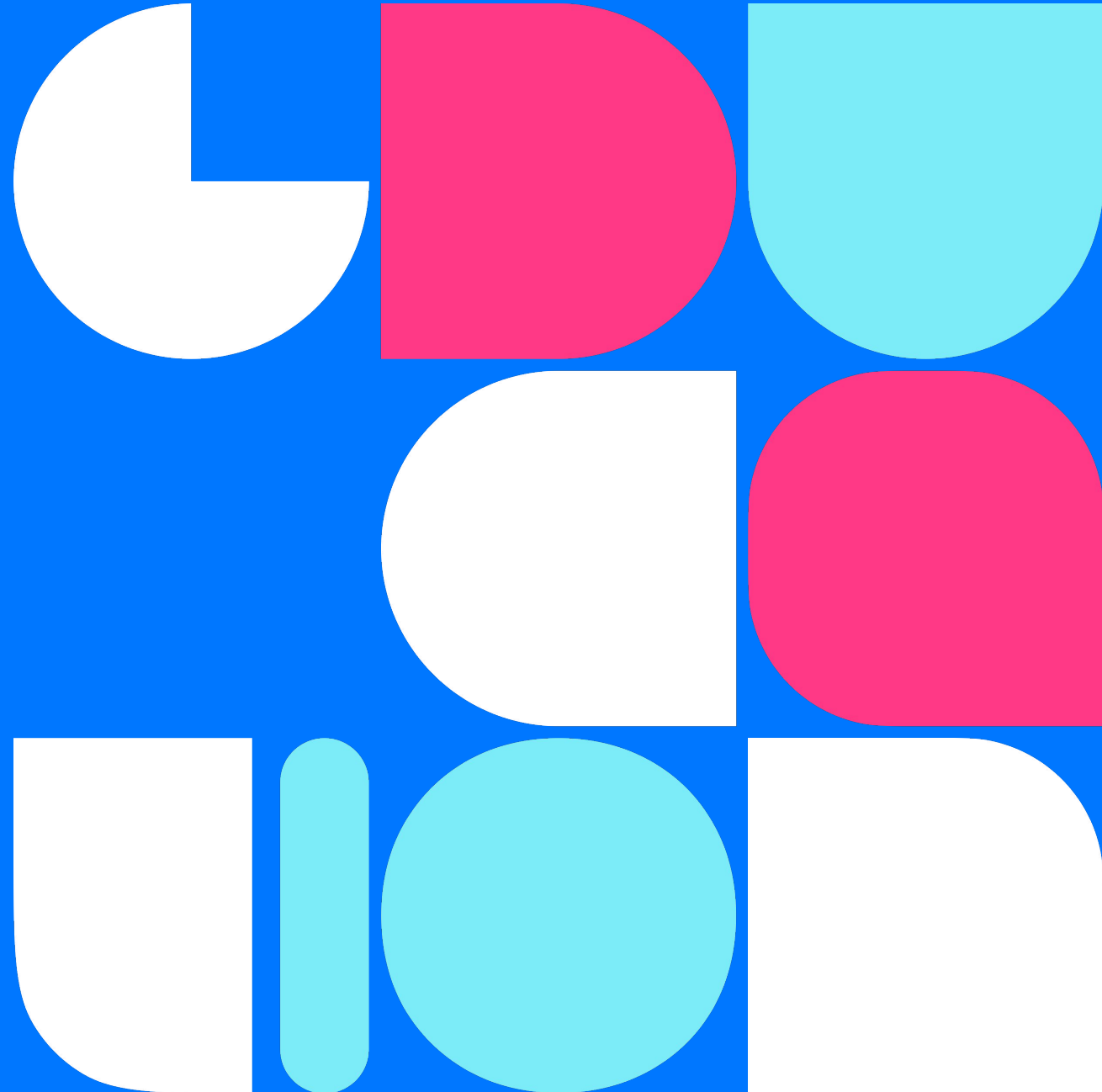
Data Pool	Dataset Source	Sample Size
X_{sota}	ShareGPT	58 K
	UltraChat	105 K
	WizardLM	143 K
X_{base}	Alpaca	52 K
	Dolly	15 K
	OAssit	10 K
	FLAN 2022	23 K

Table 1: Statistics of data pools X_{sota} and X_{base} . The Dataset Source indicates the source of the data used for sampling. The Sample Size represents the number of samples in the respective dataset.

Model	X_{sota}	X_{base}
Random Selection	5.84	4.93
Instruction Length	5.89	4.00
Perplexity	4.06	1.89
IFD	5.91	2.46
Instag Complexity	6.18	4.98
Direct Scoring (Pool=50K)	5.16	4.87
Instruction Node (Pool=50K)	5.65	4.82
EVOL COMPLEXITY (Pool=50K)	5.73	5.29
EVOL COMPLEXITY	6.27	5.57

Table 2: MT-bench of different complexity metrics. All methods select 6K samples. “Pool=50K” denotes the data selection is conducted in a 50K-sized subset due to the cost of using ChatGPT to annotate the entire pool. We include the results of our method on the 50K data pool to make a fair comparison with baselines.

Q&A





Спасибо за внимание!

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